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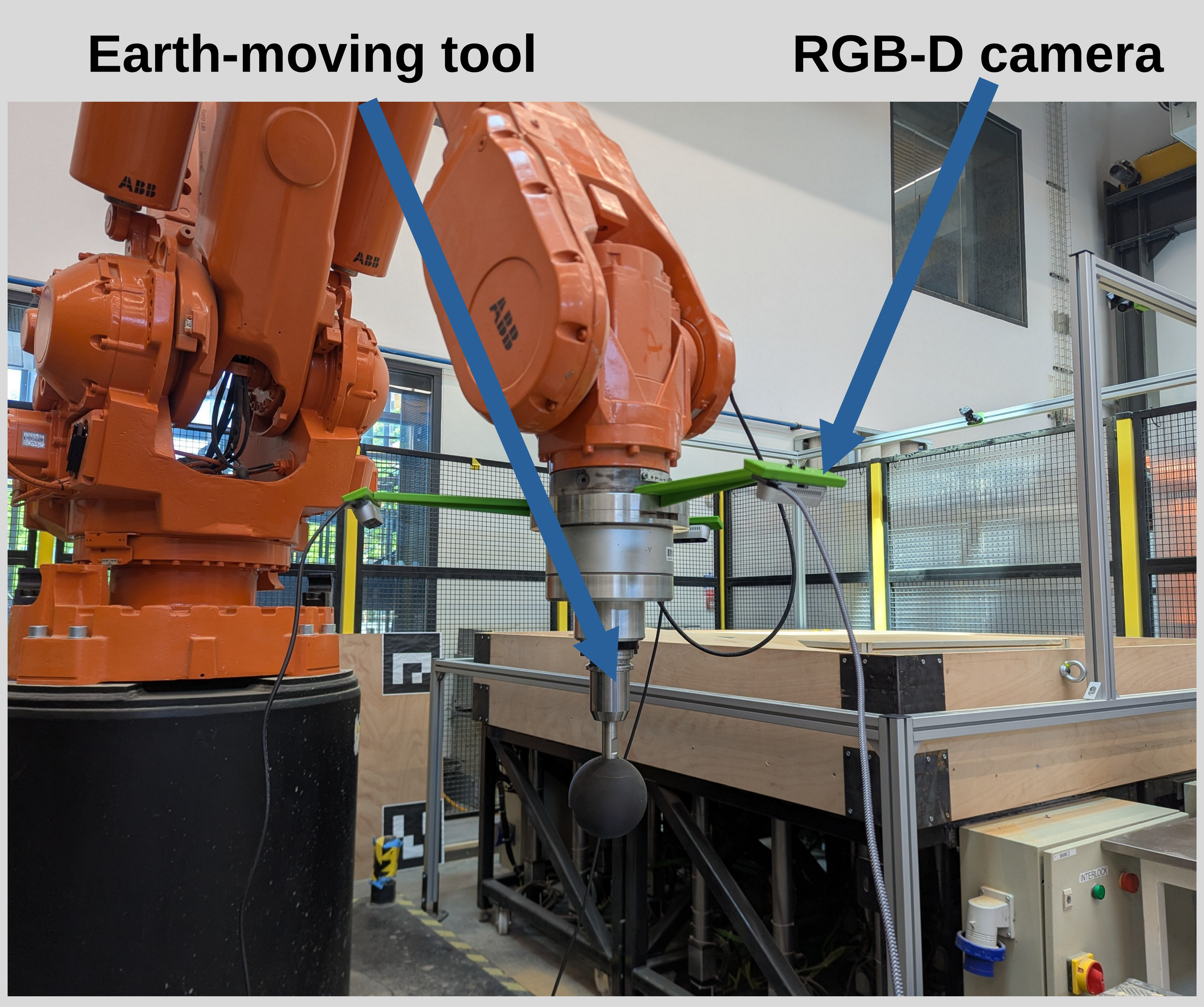
TERRA-ROBOTICS LAB

J.Ingham – come find me!

Eye-in-hand computer vision for autonomous geotechnical assessment – MaGIC 2026

Motivation

- Autonomous earth moving requires knowledge of earth state (visually)
- Lack of earth moving datasets
- Can cheap RGBD cameras replace expensive industry-standard LiDAR?
- Can point clouds be compressed and still retain geotechnical information?



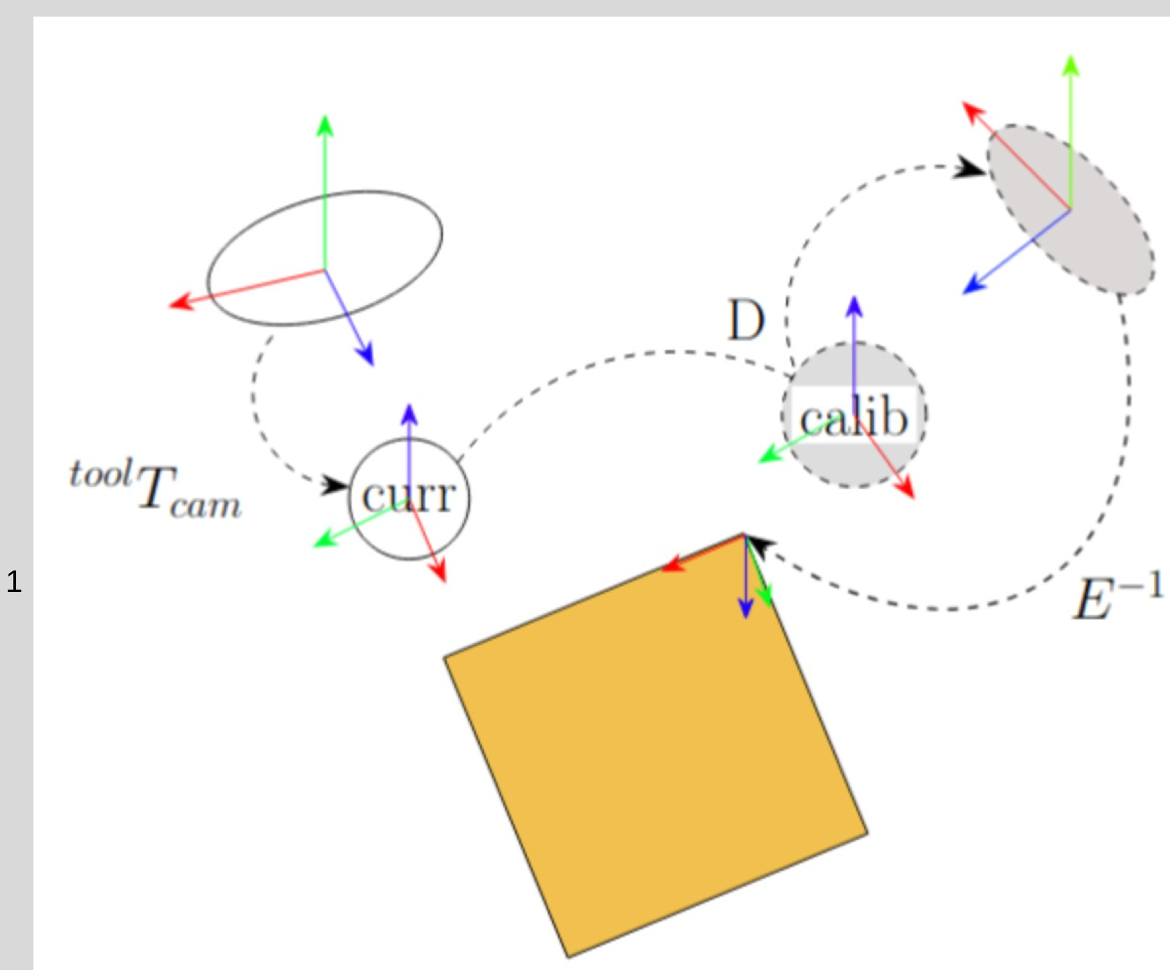
Terra Robotics Laboratory robot and multi-RGBD setup

Transformation

- RGB-D cameras capture point clouds (p_{sense}) that need to be transformed into the workspace (p_{task})
- By calibrating with a ChAruco board, the camera extrinsics E and ${}^{tool}T_{cam}$ can be calculated [1]

$$p_{task} = E^{-1} D \, {}^{tool}T_{cam} \, p_{sense}$$

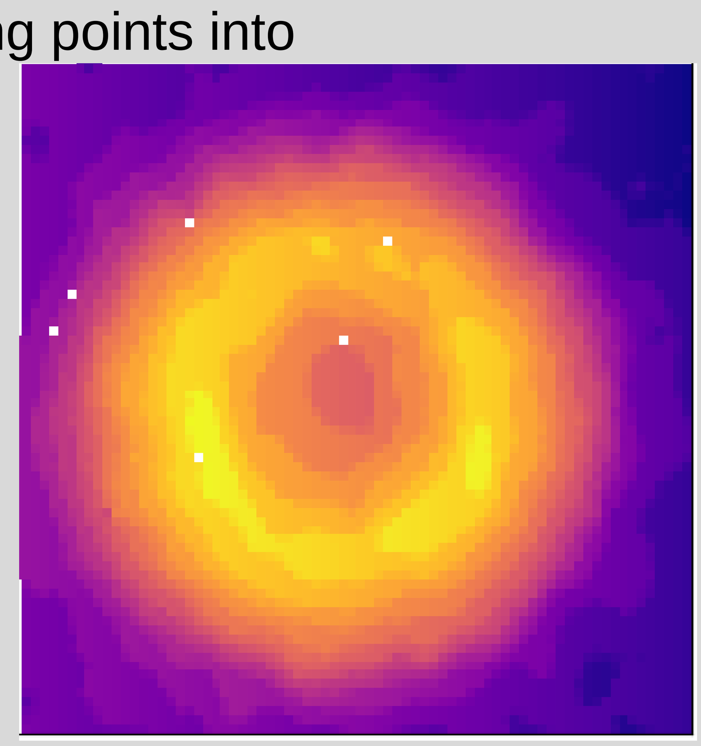
- D is the transformation from the current camera position to the calibration position



Point cloud transformation

Heightmaps

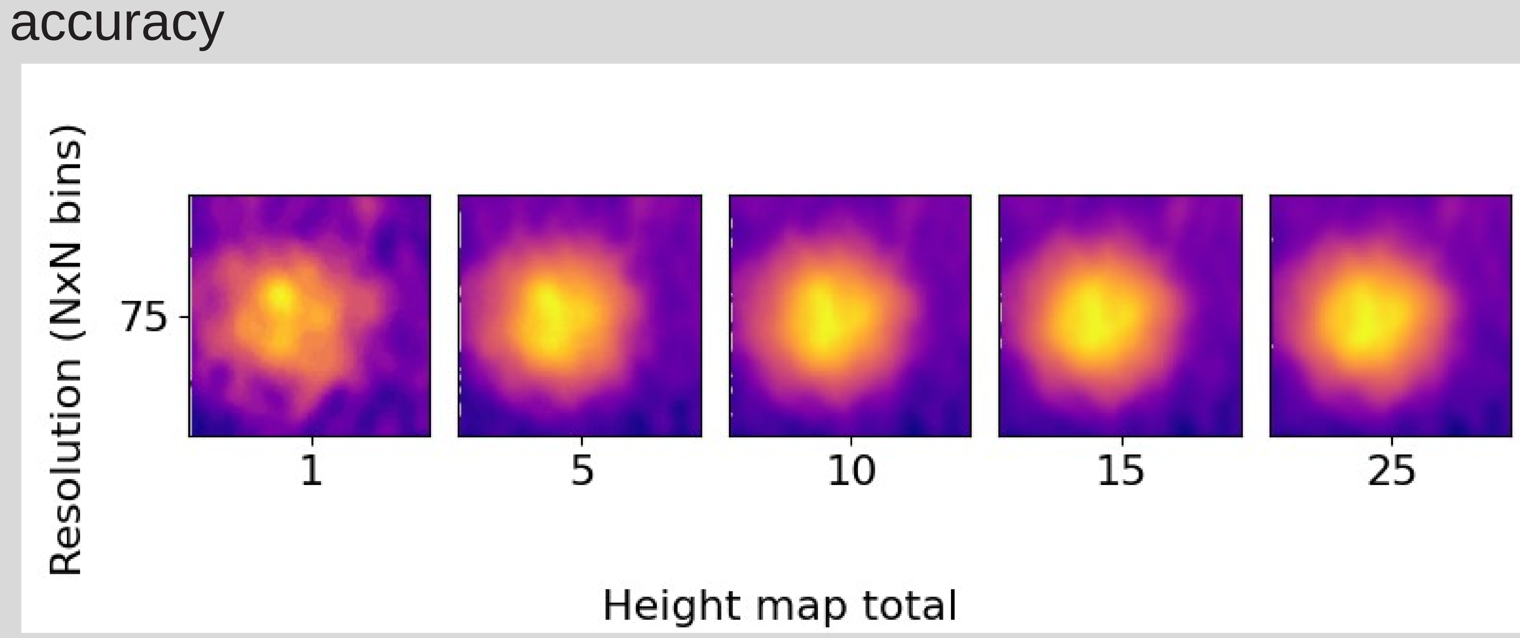
- Heightmaps are produced by vertically binning points into a grid-based format
- Turns a point cloud into an easily computable matrix
- Each cell encodes height information
- Lowers computation and memory storage requirements



An indentation transformed into a heightmap

Solo camera

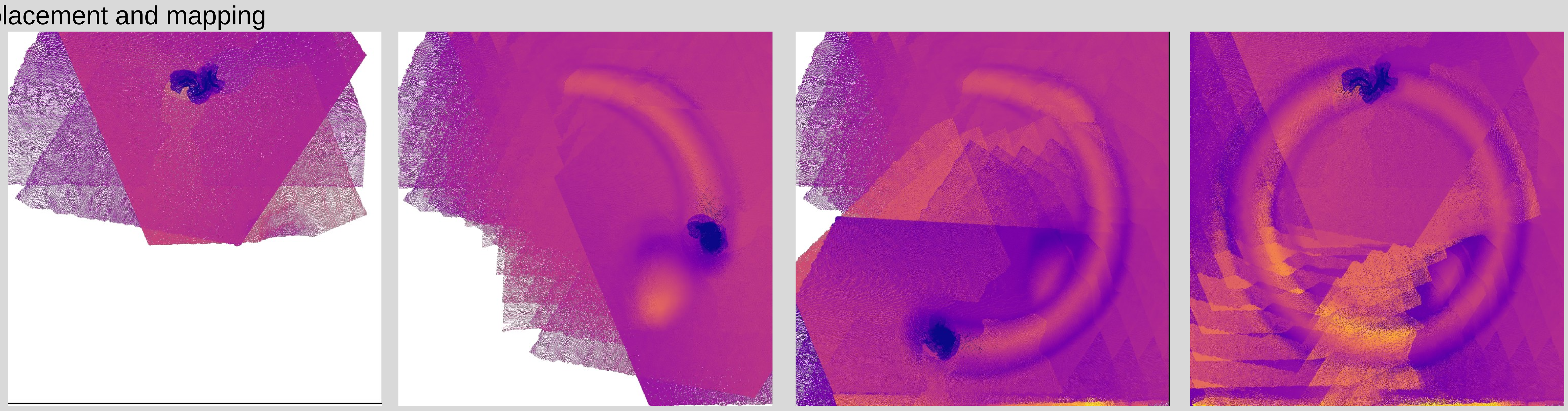
- Single camera directly above the feature
- Compared at various distances with industry standard LiDAR scanner
- At further distances, time-based heightmap averaging is required to improve accuracy



Increasingly averaged height maps

Unfiltered multiple camera results

- Simultaneous displacement and mapping
- White space represents unknown terrain
- As the tool traverses and deforms the terrain, the circular deformation can be clearly seen



Dynamic surface mapping

Future work

- Fine tuning pointcloud alignment in real-time using an Iterative Closest Point (ICP) method
- Collecting datasets which contain the force-displacement history, using a six-axis load cell
- Methodology assessment for automated earthmoving

1: E, D, and ${}^{tool}T_{cam}$ are 4x4 homogenous transformation matrices and points p_i are 4x1 column matrices

SCAN TO DISCOVER MORE

Ingham, J., Hambleton, J., Stanier, S. (2026). Terra-Robotics Laboratory: A Facility for Studying Terrain-Robot Interaction. In: Cavalcanti, A., Foster, S., Richardson, R. (eds) Towards Autonomous Robotic Systems. TAROS 2025. Lecture Notes in Computer Science(), vol 16045. Springer, Cham. https://doi.org/10.1007/978-3-032-01486-3_38